
Software Requirements Specification

for

JobVago

Version 1.0 approved

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Nanyang Technological University

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Table of Contents

Table of Contents	ii
Revision History	ii
1. Introduction	1
1.1 Purpose	1
1.2 Document Conventions	1
1.3 Intended Audience and Reading Suggestions	1
1.4 Product Scope	1
1.5 References	1
2. Overall Description	2
2.1 Product Perspective	2
2.2 Product Functions	2
2.3 User Classes and Characteristics	2
2.4 Operating Environment	2
2.5 Design and Implementation Constraints	2
2.6 User Documentation	2
2.7 Assumptions and Dependencies	3
3. External Interface Requirements	3
3.1 User Interfaces	3
3.2 Hardware Interfaces	3
3.3 Software Interfaces	3
3.4 Communications Interfaces	3
4. System Features	4
4.1 System Feature 1	4
4.2 System Feature 2 (and so on)	4
5. Other Nonfunctional Requirements	4
5.1 Performance Requirements	4
5.2 Safety Requirements	5
5.3 Security Requirements	5
5.4 Software Quality Attributes	5
5.5 Business Rules	5
6. Other Requirements	5
Appendix A: Glossary	5
Appendix B: Analysis Models	5
Appendix C: To Be Determined List	6

Revision History

Name	Date	Reason For Changes	Version
JobVago Team	April 2026	Initial SRS document	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for JobVago, a web-based application designed to assist Singaporean job seekers in navigating the employment landscape. This document covers the complete scope of JobVago Version 1.0 and is intended to guide the design, implementation, and testing phases of the project. It was produced as part of the NTU SC2006 Software Engineering course.

1.2 Document Conventions

This document follows the IEEE 830 SRS template conventions. The following typographical conventions apply:

1. Bold text indicates key terms, requirement IDs, or section headings.
2. Requirements are identified with unique tags in the format REQ-XX-NNN (e.g., REQ-AUTH-001).
3. Priority levels are stated as High, Medium, or Low.
4. "TBD" denotes information yet to be determined.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for the following audiences:

Developers: Sections 3 (External Interfaces) and 4 (System Features) provide the core specifications for implementation.

Testers / QA: Section 4 (Functional Requirements) guides test case design.

Project Managers: Sections 1 and 2 provide scope, objectives, and constraints.

Course Supervisors / Assessors: The entire document reflects the software engineering process applied.

It is recommended to read sections in order: Introduction → Overall Description → External Interface Requirements → System Features → Non-functional Requirements.

1.4 Product Scope

JobVago is a web application that aggregates Singapore Government job vacancy data (from the MyCareersFuture API on data.gov.sg) and SkillsFuture course offerings, presenting them in a unified, user-friendly interface. The application allows job seekers to browse and search for job vacancies filtered by industry, discover and bookmark relevant SkillsFuture training courses, upload and manage their resume with AI-assisted analysis and grading, and view personalised dashboards showing career statistics and recommendations. JobVago's goal is to reduce the friction Singaporean job seekers face when transitioning between jobs by connecting employment opportunities with relevant upskilling resources in a single platform.

1.5 References

MyCareersFuture API – data.gov.sg: <https://data.gov.sg>
SkillsFuture Singapore (SSG) Course API: <https://developer.ssg-wsg.gov.sg>
IEEE Std 830-1998: *IEEE Recommended Practice for Software Requirements Specifications*
NTU SC2006 Lab Manuals (Lab 1–5), AY2025/2026
DeepSeek / Ollama LLM API for AI resume analysis
VMock Resume Grading Service

2. Overall Description

2.1 Product Perspective

JobVago is a new, self-contained web application. It acts as an aggregation layer interfacing with two primary external data sources: the Singapore Government's MyCareersFuture job vacancy API (via data.gov.sg), which provides up-to-date job listings across all industries in Singapore, and the SkillsFuture Singapore (SSG) course API, which provides training and certification courses relevant to various industries. The application also integrates with AI language model services (DeepSeek via Ollama) for intelligent resume analysis and VMock for resume grading. The system follows a layered architecture: a Boundary layer (services), a Controller layer (routing and user interaction), and an Entity layer (data models).

2.2 Product Functions

The major functional areas of JobVago are:

- User Account Management: Registration, login, logout, and profile management.
- Job Vacancy Browsing: Search and filter Singapore job listings by industry, title, or keywords.
- Industry Dashboard: View career statistics including top industries and total vacancies.
- SkillsFuture Course Browsing: Browse, filter, and sort available training courses.
- Favourite Courses Management: Add or remove courses from a personal favourites list.
- Resume Management: Upload, view, edit, download, and delete resumes.
- AI Resume Analysis: Analyse uploaded resumes using an LLM to extract skills, generate feedback, and provide skill recommendations.
- Resume Grading: Grade uploaded resumes via the VMock service.
- Personal Dashboard: View past resume analyses, skill history, and course recommendations.

2.3 User Classes and Characteristics

JobVago serves two primary user classes:

2.3.1 Jobseeker (Guest / Unregistered User)

A Jobseeker is any member of the public who visits JobVago without registering. This user class can browse the career dashboard, view industry-level statistics, browse and filter job vacancies, and browse and filter SkillsFuture courses. Jobseekers cannot access personalised features such as resume management or saving favourite courses without registering.

2.3.2 Registered Jobseeker (Authenticated User)

A Registered Jobseeker has created and logged into a JobVago account. In addition to all guest capabilities, this user class can manage their personal profile and resume, upload and receive AI-powered analysis and grading of their resume, add or remove SkillsFuture courses from their favourites list, and view their personal dashboard with historical resume analyses and skill recommendations. This is the primary target audience.

2.4 Operating Environment

The software will operate in the following environment:

- Client Side: Any modern web browser (Chrome, Firefox, Safari, Edge) with JavaScript enabled.
- Server Side: A Linux-based server (Ubuntu 20.04+) running Python 3.10+ with a web framework (e.g., Flask or FastAPI).
- Database: A relational database (SQLite for development, PostgreSQL for production) for storing user data, resume analyses, and favourite courses.
- External APIs: Internet access is required to query the MyCareersFuture API and the SkillsFuture course API.
- AI Services: Access to DeepSeek or a locally-hosted Ollama LLM instance for resume text analysis.

2.5 Design and Implementation Constraints

- The system must use publicly available Singapore government APIs (data.gov.sg) and comply with their terms of use.
- AI-based resume analysis must use DeepSeek or Ollama; commercial paid alternatives are to be avoided per course guidelines.
- The back-end must be implemented in Python.
- All source code must be maintained in a GitHub repository.
- The application must handle API downtime gracefully by caching downloaded data locally.
- Resume uploads must be restricted to PDF format only.
- The system must follow the MVC-inspired layered architecture (Boundary, Controller, Entity layers) defined in the class diagram.

2.6 User Documentation

The following documentation will be delivered alongside the software:

- In-application tooltips and contextual help messages on key features.
- A README.md file in the GitHub repository describing how to install, configure, and run the application.
- This SRS document as the primary specification reference.

2.7 Assumptions and Dependencies

- The MyCareersFuture API and SkillsFuture SSG API are assumed to remain publicly available and maintain their current response formats throughout the project.
- Users are assumed to have access to a stable internet connection for API-dependent features.
- The DeepSeek / Ollama LLM service is assumed to produce resume analysis of sufficient quality for the application's purposes.
- VMock API integration is assumed to remain available; if access is lost, this feature will be degraded gracefully.
- The project is developed for demonstration and academic purposes and is not expected to handle production-scale traffic.

3. External Interface Requirements

3.1 User Interfaces

The application shall provide a responsive web-based user interface with the following characteristics:

- A navigation bar on all pages with links to: Home/Dashboard, Browse Jobs, Browse Courses, Resume Manager, and Login/Register or Logout.
- A career dashboard page displaying total job vacancies, top industries, and total SkillsFuture courses available.
- A job vacancy listing page with search and filter controls (by industry, job title, keywords).
- A SkillsFuture course browsing page with filter controls (by category, provider, language, subsidy, skill level) and a favouriting button on each course card.
- A resume management page allowing file upload (drag-and-drop or click), with buttons to Edit, Analyse, Grade, Download, and Delete the current resume.
- A personal dashboard page showing the user's skill recommendations, analysis history, and favourite courses.
- Error messages displayed inline and in user-friendly language.
- The UI shall be usable on standard desktop resolutions (1280x720 and above).

3.2 Hardware Interfaces

JobVago is a web-based application and does not interface directly with any hardware. Users access the application through standard computing devices (PC, laptop, or tablet) equipped with a web browser.

3.3 Software Interfaces

The following external software interfaces are required:

3.3.1 MyCareersFuture API (data.gov.sg)

Purpose: Fetch Singapore job vacancy listings in CSV format.

Interface Type: REST API / CSV download endpoint.

Data: Job title, employer, industry, salary, location, date posted.

Refresh Strategy: Data is downloaded and cached at application startup with a configurable timeout.

3.3.2 SkillsFuture Singapore (SSG) Course API

Purpose: Fetch available SkillsFuture training course listings.

Interface Type: REST API / CSV download.

Data: Course title, provider, category, language, price, subsidised price, skill tags, prerequisites.

3.3.3 DeepSeek / Ollama LLM API

Purpose: AI-powered resume analysis – extract skills, produce feedback, and generate improvement suggestions.

Interface Type: REST API (local or remote).

Data In: Resume text extracted from uploaded PDF. Data Out: Skills list, feedback text, grade scores, action verbs.

3.3.4 VMock Resume Grading Service

Purpose: AI-based resume grading producing quantitative scores (overall, content, presentation).

Interface Type: REST API. Data In: Resume PDF. Data Out: Grade scores and rule-based feedback.

3.3.5 Relational Database

- Persists User accounts, UserSkill records, FavouriteCourse records, ResumeAnalysis results, and DashboardStats.
- SQLite is used in development; PostgreSQL is recommended for deployment.

3.4 Communications Interfaces

- All communication between the web browser and the back-end server shall use HTTPS (TLS 1.2 or higher) in production.
- External API calls are made server-side over HTTPS.
- Standard HTTP status codes are used to communicate success (2xx) and error (4xx/5xx) states.
- User authentication is managed via secure, httpOnly session cookies or JWT tokens.

4. System Features

The following system features correspond to the use cases identified in the JobVago Use Case Diagram.

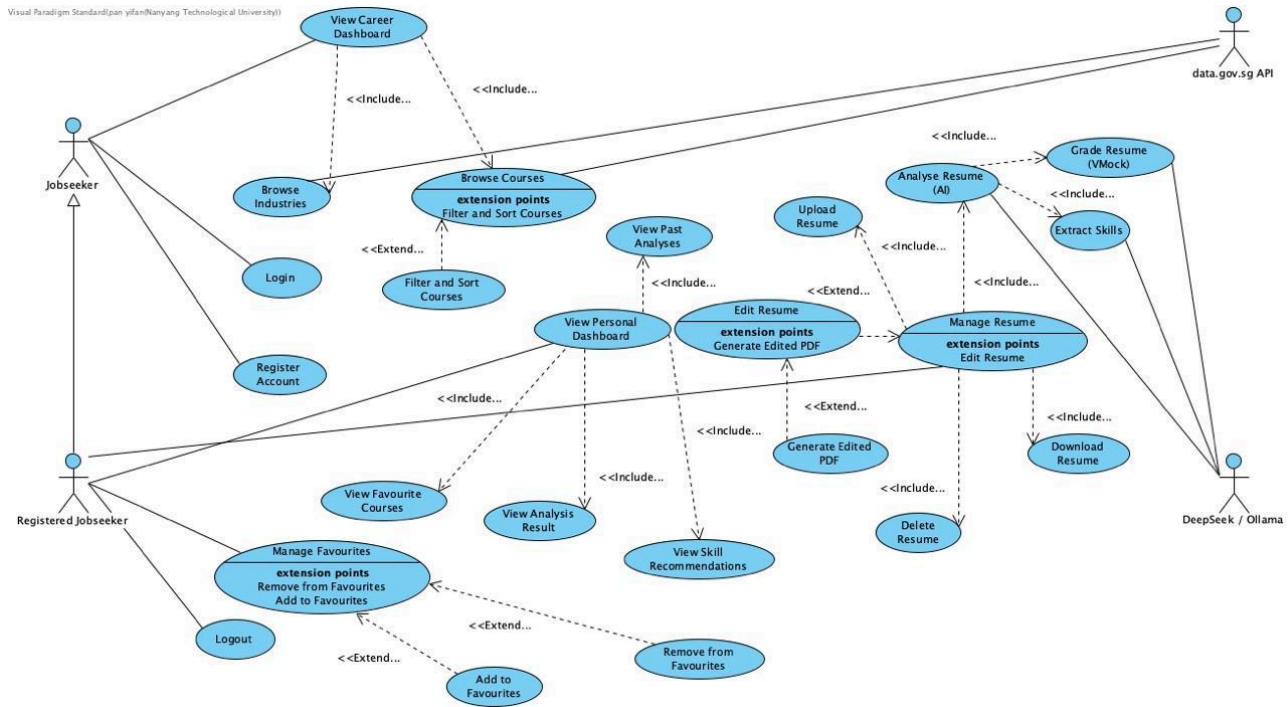


Fig 1.UseCase Diagram

4.1 User Registration and Authentication

4.1.1 Description and Priority

Allows users to create an account, log in, and log out of JobVago. This feature gates access to all personalised functionality. Priority: High.

4.1.2 Stimulus/Response Sequences

Register: User fills in full name, email, and password → System validates uniqueness of email → System creates account and redirects to dashboard.

Login: User submits email and password → System authenticates credentials → System creates session and redirects to personal dashboard.

Logout: User clicks Logout → System destroys session → User is redirected to homepage.

4.1.3 Functional Requirements

REQ-AUTH-001: The system shall allow a user to register with a unique email address, full name, and password.

REQ-AUTH-002: The system shall hash passwords before storing them in the database.

REQ-AUTH-003: The system shall validate that the email format is correct during registration.

REQ-AUTH-004: The system shall reject login attempts with incorrect credentials and display a generic error message.

REQ-AUTH-005: The system shall maintain user sessions across page refreshes.

REQ-AUTH-006: The system shall allow authenticated users to log out, which must invalidate the current session.

4.2 View Career Dashboard

4.2.1 Description and Priority

Provides all users (registered and guest) with a high-level overview of the Singapore job market, including total vacancies, top industries by job count, and total SkillsFuture courses available.

Priority: High.

4.2.2 Stimulus/Response Sequences

User navigates to the homepage → System retrieves DashboardStats from DataService → System renders statistics, top industries, and industry growth indicators.

4.2.3 Functional Requirements

REQ-DASH-001: The system shall display total job vacancies, total number of industries, and total SkillsFuture courses on the career dashboard.

REQ-DASH-002: The system shall display the top industries ranked by number of job vacancies.

REQ-DASH-003: The system shall display a growth rate indicator for each listed industry.

REQ-DASH-004: Dashboard data shall be sourced from downloaded datasets and refreshed at application startup.

4.3 Browse Industries and Job Vacancies

4.3.1 Description and Priority

Allows all users to browse Singapore job vacancies and filter by industry. Priority: High.

4.3.2 Stimulus/Response Sequences

User selects an industry → System fetches job listings for that industry → System renders a list of vacancies with title, employer, salary range, and date.

User applies filters → System re-filters results and updates the display.

4.3.3 Functional Requirements

REQ-JOB-001: The system shall display a list of job vacancies sourced from the MyCareersFuture API dataset.

REQ-JOB-002: The system shall allow users to filter job listings by industry.

REQ-JOB-003: The system shall display for each vacancy: job title, employer name, salary range (if available), location, and posting date.

REQ-JOB-004: The system shall allow users to download the full job vacancies dataset as a CSV file.

REQ-JOB-005: The system shall display industry-level statistics (vacancy count, icon, growth indicator) on the industry detail page.

4.4 Browse and Filter SkillsFuture Courses

4.4.1 Description and Priority

Allows all users to browse available SkillsFuture courses and apply filters. Registered users may add courses to their favourites. Priority: High.

4.4.2 Stimulus/Response Sequences

User navigates to the Courses page → System fetches course data → System renders course cards with title, provider, price, and subsidy information.

User selects filters → System applies filters and updates the displayed list.

Registered user clicks Add to Favourites → System saves the FavouriteCourse record linked to the user.

4.4.3 Functional Requirements

REQ-CRS-001: The system shall display a list of SkillsFuture courses fetched from the SSG course dataset.

REQ-CRS-002: The system shall allow users to filter courses by industry, provider, language, training mode, impact level, and subsidy eligibility.

REQ-CRS-003: The system shall display for each course: title, provider, full price, subsidised price, language, mode, and skill tags.

REQ-CRS-004: Registered users shall be able to add a course to their favourites list.

REQ-CRS-005: Registered users shall be able to remove a course from their favourites list.

REQ-CRS-006: The system shall display the user's favourited courses on their personal dashboard.

4.5 Resume Management

4.5.1 Description and Priority

Allows registered users to upload, view, edit, download, and delete their resume. This feature is the gateway to AI analysis and grading. Priority: High.

4.5.2 Stimulus/Response Sequences

Upload: Registered user selects a resume file → System validates the file type → System saves the file → System confirms successful upload.

Edit: User navigates to resume editor → System loads current resume data → User modifies content → System saves changes and optionally regenerates a PDF.

Download: User clicks Download Resume → System serves the resume PDF to the browser.

Delete: User clicks Delete Resume → System confirms with the user → System removes the resume file and clears the path from the user record.

4.5.3 Functional Requirements

REQ-RES-001: The system shall allow registered users to upload a resume file in PDF format.

REQ-RES-002: The system shall validate uploaded files and reject disallowed file types with an appropriate error message.

REQ-RES-003: The system shall store the path to the user's most recent resume in the user's profile record.

REQ-RES-004: The system shall allow the user to download their currently stored resume.

REQ-RES-005: The system shall allow the user to edit resume fields through a structured editor and regenerate a PDF from the updated data.

REQ-RES-006: The system shall allow the user to delete their stored resume, removing both the file and the database record.

REQ-RES-007: The system shall indicate whether a user currently has a resume on file.

4.6 AI Resume Analysis

4.6.1 Description and Priority

Allows registered users to submit their uploaded resume for AI-powered analysis. The system extracts skills, evaluates quality, and provides actionable feedback and SkillsFuture course recommendations. Priority: High.

4.6.2 Stimulus/Response Sequences

User clicks Analyse Resume → System confirms a resume is uploaded → System sends resume text to the AI model → System receives results → System saves the ResumeAnalysis record → System displays analysis results including skills, scores, and recommendations.

User clicks Grade Resume (VMock) → System submits resume to VMock → System displays grade scores and rule-based feedback.

User navigates to View Past Analyses → System retrieves all ResumeAnalysis records for the user → System displays history in reverse chronological order.

4.6.3 Functional Requirements

REQ-AI-001: The system shall extract and display a list of skills identified from the user's resume.

REQ-AI-002: The system shall produce and display an overall grade score, a content score, and a presentation score for the resume.

REQ-AI-003: The system shall identify and display action verbs found in the resume.

REQ-AI-004: The system shall produce textual feedback on the resume (areas of strength and areas for improvement).

REQ-AI-005: The system shall save each resume analysis result as a ResumeAnalysis record linked to the user, including timestamp and scores.

REQ-AI-006: The system shall display a breakdown of scores (content, presentation, total) in the analysis result view.

REQ-AI-007: The system shall recommend relevant SkillsFuture courses based on skills identified or lacking in the resume.

REQ-AI-008: The system shall allow the user to view a history of all their past resume analyses.

4.7 Personal Dashboard

4.7.1 Description and Priority

Provides registered users with a centralised personal overview: their latest resume analysis, skill profile, favourite courses, and personalised course recommendations. Priority: Medium.

4.7.2 Stimulus/Response Sequences

User navigates to Personal Dashboard → System retrieves the user's latest analysis, UserSkill records, and FavouriteCourse records → System renders the dashboard with all personalised data.

4.7.3 Functional Requirements

REQ-PDASH-001: The system shall display the user's current skill profile (skills extracted from resume analyses).

REQ-PDASH-002: The system shall display the user's favourite courses on the personal dashboard.

REQ-PDASH-003: The system shall display a summary of the most recent resume analysis result.

REQ-PDASH-004: The system shall display the total number of resume analyses the user has completed.

5. Other Non-functional Requirements

5.1 Performance Requirements

The career dashboard shall load within 3 seconds under normal network conditions.

Job vacancy and course listing pages shall render within 3 seconds once data has been cached locally.

Resume analysis (AI processing) is permitted to take up to 60 seconds; the UI shall display a loading indicator while analysis is in progress.

The system shall support at least 30 concurrent users without significant degradation in response time.

5.2 Safety Requirements

There are no physical safety concerns associated with this application. The system must ensure data integrity: user data (resume, analysis history) shall not be permanently deleted without explicit user confirmation, and database transactions shall be atomic to prevent partial writes in the event of failure.

5.3 Security Requirements

- User passwords shall be hashed using a strong algorithm (e.g., bcrypt) before storage.
- All authenticated routes shall verify the user's session before returning data.
- Users shall only be able to access their own resume files and analysis records (server-side access control enforced).
- Input fields shall be protected against SQL injection and XSS attacks.
- The system shall use HTTPS in any production or demo deployment.

5.4 Software Quality Attributes

Usability: The interface shall be intuitive for non-technical users. Key actions shall be accessible within 2 clicks from the homepage.

Maintainability: The codebase shall follow the layered architecture (Boundary / Controller / Entity), making it straightforward to swap external APIs or add features.

Reliability: The system shall handle external API unavailability gracefully by using cached data and displaying informative messages to the user.

Testability: All control classes shall expose clearly defined public methods to facilitate unit testing using equivalence class, boundary value, and basis path techniques.

Portability: The application shall be deployable on any standard Linux server with Python 3.10+ and shall not depend on a specific operating system.

5.5 Business Rules

- Only registered and authenticated users may upload, edit, analyse, or delete resumes.
- Only registered and authenticated users may save or remove favourite courses.
- A user may only have one active resume on file at a time; uploading a new resume replaces the previous one.
- A user may add a specific course to their favourites only once; duplicate favourites are not permitted.
- Guest users may browse all public data (jobs, courses, career dashboard) without restriction.

6. Other Requirements

The application must be deployable and demonstrable in the NTU SC2006 lab environment.

All source code, test cases, and documentation (including this SRS) must be submitted to the team's GitHub repository by the deadline specified in the Lab 5 manual.

The application shall support the English language only (internationalisation is out of scope for Version 1.0).

7. Appendix A: Glossary

API: Application Programming Interface – a defined interface that allows software components to communicate.

CSV: Comma-Separated Values – a plain-text file format used to transfer tabular data.

data.gov.sg: Singapore Government's open data portal, hosting the MyCareersFuture job vacancy dataset.

DeepSeek / Ollama: Open-source large language model (LLM) services used for AI-powered resume analysis in JobVago.

FavouriteCourse: An entity representing a SkillsFuture course that a registered user has bookmarked.

JobVago: The name of the web application specified in this document.

LLM: Large Language Model – an AI model capable of text understanding and generation.

PDF: Portable Document Format – the file format used for resume uploads and downloads.

ResumeAnalysis: An entity recording the results of one AI analysis session on a user's resume.

SkillsFuture: A national programme by the Singapore Government encouraging lifelong learning and upskilling.

SRS: Software Requirements Specification – this document.

SSG: SkillsFuture Singapore – the government agency administering the SkillsFuture programme.

TBD: To Be Determined – used as a placeholder where information is not yet available.

UserSkill: An entity representing a skill associated with a user, extracted from resume analysis.

VMock: An AI-based resume evaluation service used for grading resumes in JobVago.

8. Appendix B: Analysis Models

The following design models have been produced for JobVago and are submitted as separate artefacts alongside this SRS:

- Use Case Diagram – shows interactions between Jobseeker, Registered Jobseeker, data.gov.sg API, and DeepSeek/Ollama actors with the system's use cases.
- Class Diagram – shows the Boundary (service), Controller, and Entity layers of the JobVago system, including DataDownloader, DataService, CourseService, IndustryService, AuthenticationService, ResumeEditorService, ResumeAnalysisService, SkillExtractionService, ResumeGradingService, FavouriteCourseService, and all entity classes.
- UI State Machine / Dialog Map – describes the navigation flow between application screens.
- Sequence Diagrams – detailed interaction diagrams for key use cases (to be finalised with implementation).

9. Appendix C: To Be Determined List

1. Exact SkillsFuture SSG API endpoint and authentication method – to be confirmed during implementation.
2. VMock API availability and access credentials for production – fallback: display message if VMock is unavailable.
3. Sequence diagrams for all use cases – to be completed and submitted with Lab 5 deliverables.
4. UI mockups / wireframes for all screens – to be attached to the UI state machine document.